Family name:_____ Given name: _____

Exam

Advanced Topics in Engineering Physics ('Plasmas')

Summer term 2022

Constants (approximated values):

$e = 1.60 \cdot 10^{-19}$ A·s $h = 6.63 \cdot 10^{-34}$ Js $k_B = 1.38 \cdot 10^{-23}$ m²·kg·s^{-2}·K^{-1}
$m_e = 9.11 \cdot 10^{-31}$ kg $\varepsilon_0 = 8.85 \cdot 10^{-12}$ A·s·V^{-1}·m^{-1} $c = 3 \cdot 10^8$ m/s 1 eV = $1.6 \cdot 10^{-19}$ J

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1. Please indicate whether the following statements are true or false by checking the appropriate box (Note: Only one box can be checked and only right answers are considered whereas wrong ones are disregarded, so there are no 'negative points' possible.)

Statement	**True**	**False**
Plasmas are always fully ionised.		**X**
The dissociation energy of a gas is notably lower than its ionisation energy.	**X**	
Plasmas are quasi neutral; there is no charge on a macroscopic scale.	**X**	
The term 'Debye length' is another expression for the length of a plasma.		**X**
In an equilibrium plasma, the electron temperature is lower than both the ion temperature and neutral gas temperature.		**X**
The Townsend discharge is a self-maintaining discharge.		**X**

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2. A plasma source consist of two plane-parallel circular metallic electrodes. The diameter of the electrodes is D = 26 mm and the distance (= air gap thickness; resistor of air R = $1.5 \cdot 10^{14}$ Ω) is d = 5 cm. The voltage applied to the electrodes is V = 150 V.

a. Determine the electric field E between the electrodes.
b. Determine the current density J between the electrodes.
c. Determine the drift velocity v_{Drift} of charged particles within this plasma where the particle's mobility amounts to 74 A·s²/kg.

a)
U = E x d → E = U/d = 150 V/0.05 m ≈ 3 kV/m

b)
J = I/A where

- **I = U/R = 150 V/1.5 x 10^{14} Ω = 1 x 10^{-12} A**
  and
- **A = (D/2)² x π ≈ 5.3 x 10^{-4} m²**

→ J = 1 x 10^{-12} A/5.3 x 10^{-4} m² ≈ 1.9 x 10^{-9} A/m²

c)
v_{Drift} = μ x E = 74 A·s²/kg x 3 kV/m = 222000 m/s

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3. Determine the

a. electron density n_e and
b. the plasma frequency ω_p

of a plasma in thermodynamic equilibrium where the following values are known:

- Plasma temperature T = 11604K
- Debye radius λ_D = 235 µm.

a)
10^{15}/m², since

$$\lambda_D = \sqrt{\frac{8.85 \cdot 10^{-12} \frac{As}{Vm} \cdot 1.38 \cdot 10^{-23} \frac{m^2 \cdot kg}{s^2 \cdot K} \cdot 11604K}{\frac{10^{15}}{m^3} \cdot (1.6 \cdot 10^{-19} As)^2}} = 2.35 \cdot 10^{-4} m \approx 235 \mu m.$$

b)

$$\omega_p = \sqrt{\frac{(1,60218 \cdot 10^{-19} As)^2 \cdot \frac{10^{15}}{m^3}}{9,10938 \cdot 10^{-31} kg \cdot 8,85419 \cdot 10^{-12} \frac{As}{Vm}}} = 1.783.990.256 s^{-1} \approx 1,78 GHz$$

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4. A plasma is operated at atmospheric pressure; the used process gas is molecular nitrogen with an ionisation energy of E_{ion} = 15.7 eV. Determine the 1rst Townsend coefficient α for electrons with an energy of E_{e-} = $2 \cdot 10^{-19}$ J.

$2 \cdot 10^{-19}$ J = 1.25 eV →

$$\alpha = \frac{1}{\lambda} \cdot e^{\left(\frac{E_{ionisation}}{E_{e^-}}\right)} = \frac{1}{68 \cdot 10^{-9} m} \cdot e^{\left(\frac{15.7 eV}{1.25 eV}\right)} = 4.2 \cdot 10^{12} / m$$

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5. A plasma is ignited using argon as process gas (ionisation energy E_{ion} = 15.75 eV, mass m_{Ar} = 6.63·10^{-26} kg). Within this plasma, an electron is generated via photoelectric emission at a copper electrode surface. The work function of this surface is W = 4.55 eV and the wavelength of the incident light is 70 nm.

a) Does the generated electron provide sufficient kinetic energy for the ionisation of argon (neglecting a possible acceleration of the electron by the electric field)?
b) Determine both the maximum energy E (given in eV) and temperature T (given in K) of argon ions within the plasma assuming an electron energy as determined in a). (hint: 1 eV = 11604K)
c) Determine the potential (given in eV) within the sheath of this plasma.

a)
Electron energy = 17.712 eV – 4.55 eV = 13.162 eV > not sufficient, since 70 nm corresponds to 17.86 eV

b)

$$\frac{v_e}{v_i} > \sqrt{\frac{m_i}{m_e}} > \sqrt{\frac{6,63391 \cdot 10^{-26} kg}{9,10938 \cdot 10^{-31} kg}} \approx 270$$

The ion velocity is thus v_e/270 = <u>48.75 x 10^{-3} eV</u>
The ion temperature thus amounts to 48.75 x 10^{-3} eV x 11604K = <u>565.7K</u>

c)

$$\Phi \approx -13.162 eV \cdot ln \sqrt{\frac{6,63391 \cdot 10^{-26} kg}{2 \cdot \pi \cdot 9,10938 \cdot 10^{-31} kg}} = -61.58 eV.$$

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6. An atmospheric pressure plasma in perfect local thermal equilibrium is operated at a voltage of V = 17 kV where the discharge gap thickness is d = 2 mm. Determine the energy of electrons within this plasma.

E = U/d = 17 kV/0.002 m = 8500000 V/m

→ $E_{e^-} = \lambda \cdot E = 68 \cdot 10^9$ m x 8500000 V/m = 0.578 eV.